

Liivv — Architecture Deep Dive

Audience: Product, ops, and engineering

App: BigCommerce Catalyst storefront (core/) + Liivv health and pharmacy extensions

Date: August 2026

Status: Current

Overview: [Liivv-Architecture.md](#) — product-shaped summary of the same system

PDF: [Liivv-Architecture-Deep-Dive.pdf](#) (same content, for email / print)

Reading map:

- **§1–§2** — systems of record, /bc-app, system context
- **§3–§5** — data flows (commerce, onboarding/pharmacy, care chat)
- **§6** — security controls (S1–S8)
- **§7–§9** — auth, trust boundary, secrets, evidence paths

Assumes the [overview](#): two engines (BigCommerce shop vs Canadian Supabase health records), Next.js on Vercel as the trust boundary, Stripe for cards, no Rx photo uploads.

1. System of record

Domain	System of record	Notes
Products, categories, prices	BigCommerce	Synced to Stripe prices via BC webhooks
Customers (login identity)	BigCommerce	Linked in Supabase profiles.bigcommerce_customer_id
Cart / checkout cart	BigCommerce GraphQL	Cart ID stored in signed Auth.js / anonymous JWT
Orders	BigCommerce	Created via Admin REST after Stripe success
Payment methods & subscriptions	Stripe	Renewals create BC orders on invoice.paid
Health profile, insurance	Supabase	Not modeled in BC
Prescriptions, CarePack, refills	Supabase	Staff queue in /bc-app
Live chat messages	Supabase	Optional OpenAI bot replies (flagged off)

Recurring orders: Stripe handles the repeating charge. When a renewal succeeds, Liivv still creates the order in BigCommerce. Stripe is the billing engine. BigCommerce remains the only order engine.

Chat assistant: Care conversations live in Supabase. If VIRTUAL_CARE_BOT_ENABLED=true and an API key is set, **customer message text is sent to OpenAI** even when the bot refuses clinical advice and points people to a pharmacist. That flag stays **false** until the team decides whether to buy an OpenAI arrangement that includes a **DPA**. Human care-team chat in /bc-app still stays in Supabase either way.

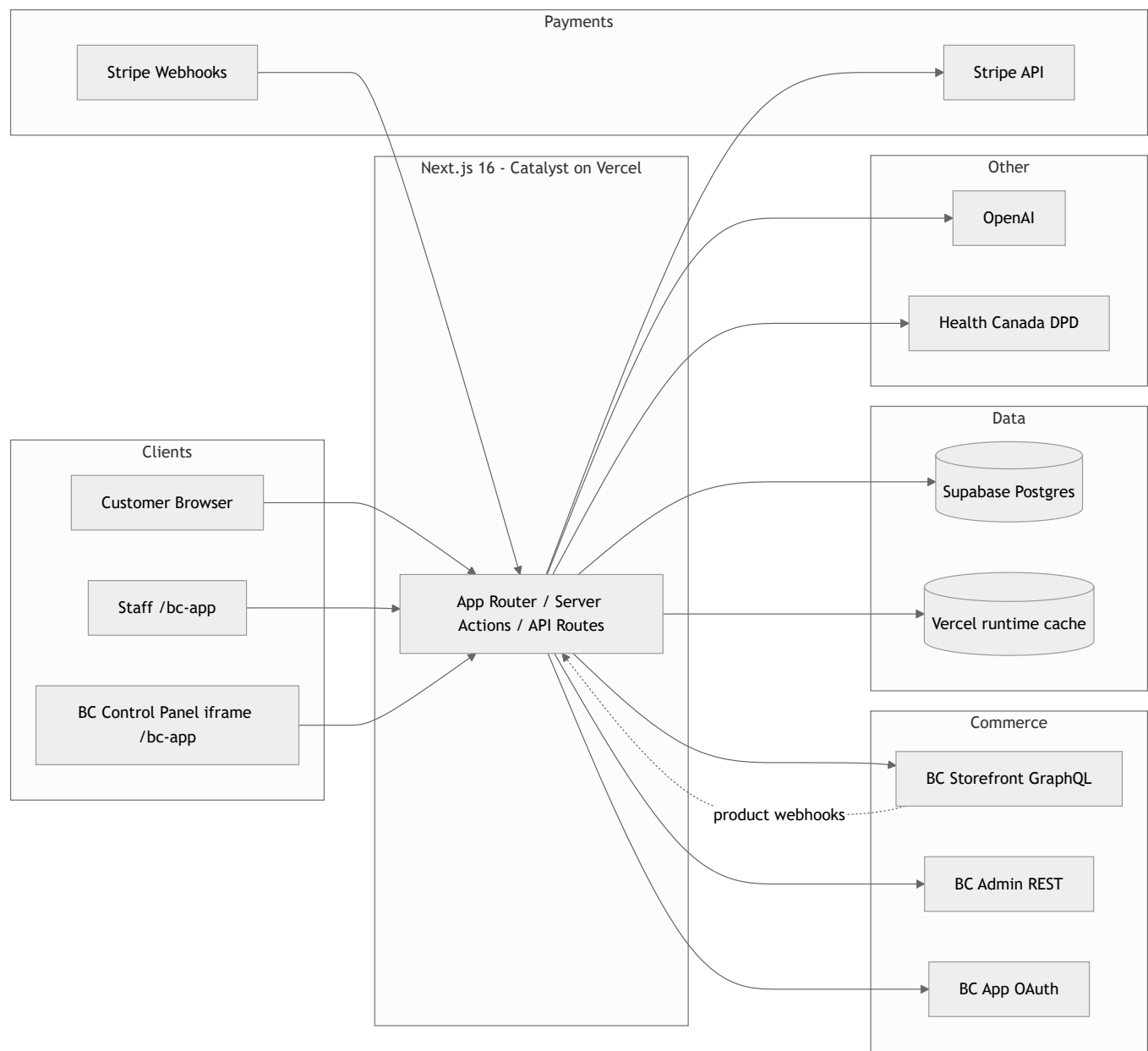
Patients do not upload prescription photos. Adding a prescription is **transfer from another pharmacy** or **doctor fax**. Staff work the queue in `/bc-app` (Liivv Staff).

`/bc-app` (**Liivv Staff**): pharmacy and care-team staff open it from the **BigCommerce control panel** → **Apps** → **My apps** → **Liivv Staff**. BigCommerce loads the app in an iframe using the **signed-in load token**. Tabs: **pharmacy** (prescription / refill / CarePack queues), **customers**, **chat**. Session cookie `liivv_bc_app` is scoped to `/bc-app` for 12 hours and bound to the store hash. `/staff` is 404. `/admin` only redirects to the BC control panel. Product-shaped summary: [overview §2](#).

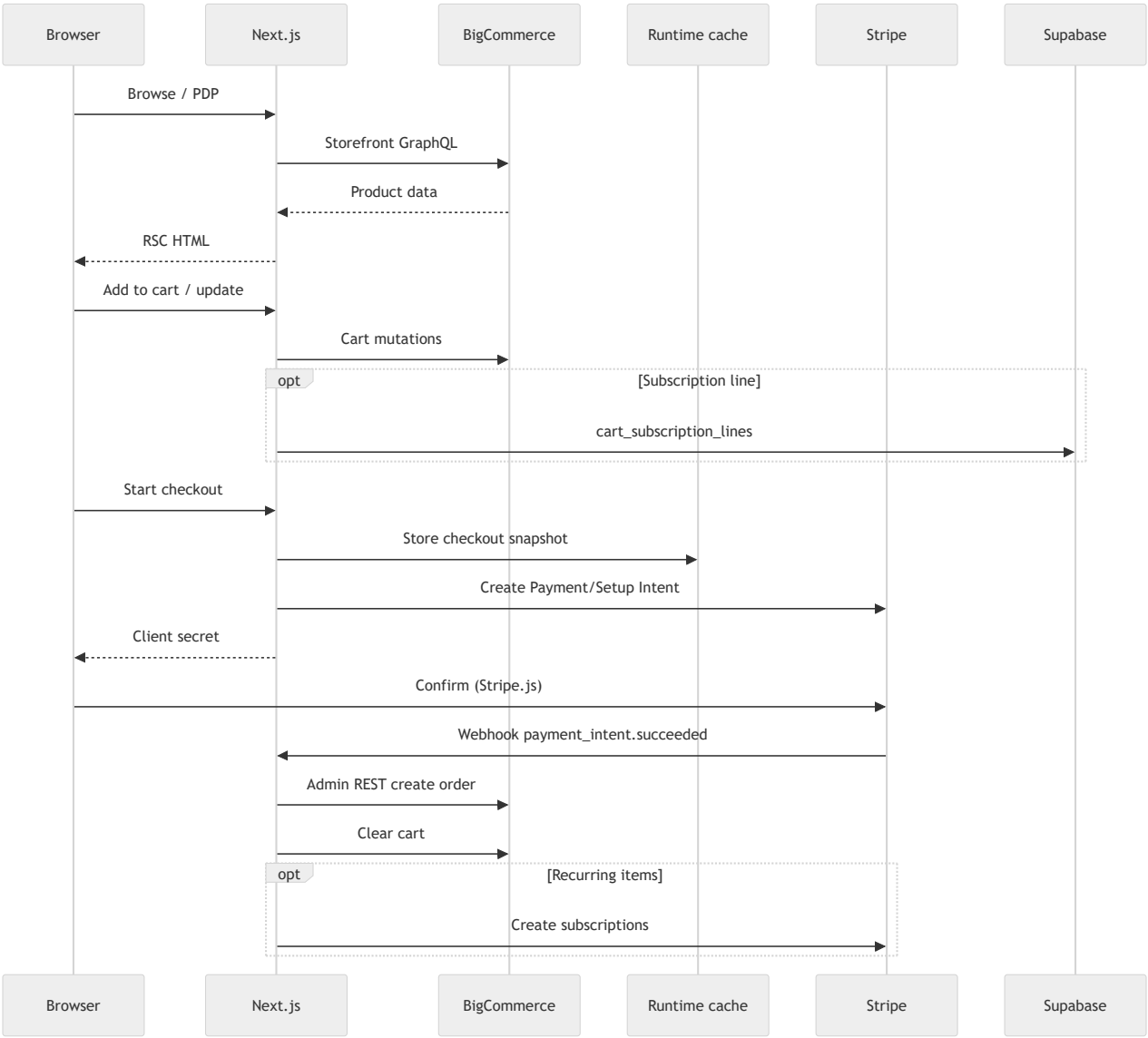
Health Canada DPD: the Drug Product Database is Health Canada's **free public API** (open government data — no key, no paid contract). Liivv proxies it so customers can search licensed Canadian drugs by brand name and add the selected product to a prescription. The **prescription is stored in Supabase**. Search queries go to Health Canada; patient records do not. The proxy is rate-limited at 60 requests / IP / minute so Liivv is not used as an open relay.

Hosting: the Catalyst app (`core/`) runs on **Vercel** (production and preview). Next.js on Vercel is the only process that holds service-role, admin, and payment secrets. Secrets live in the Vercel project (encrypted) plus gitignored `.env.local` for local work. Checkout snapshots and the staff-app install token use **Vercel's runtime cache** (in-memory locally).

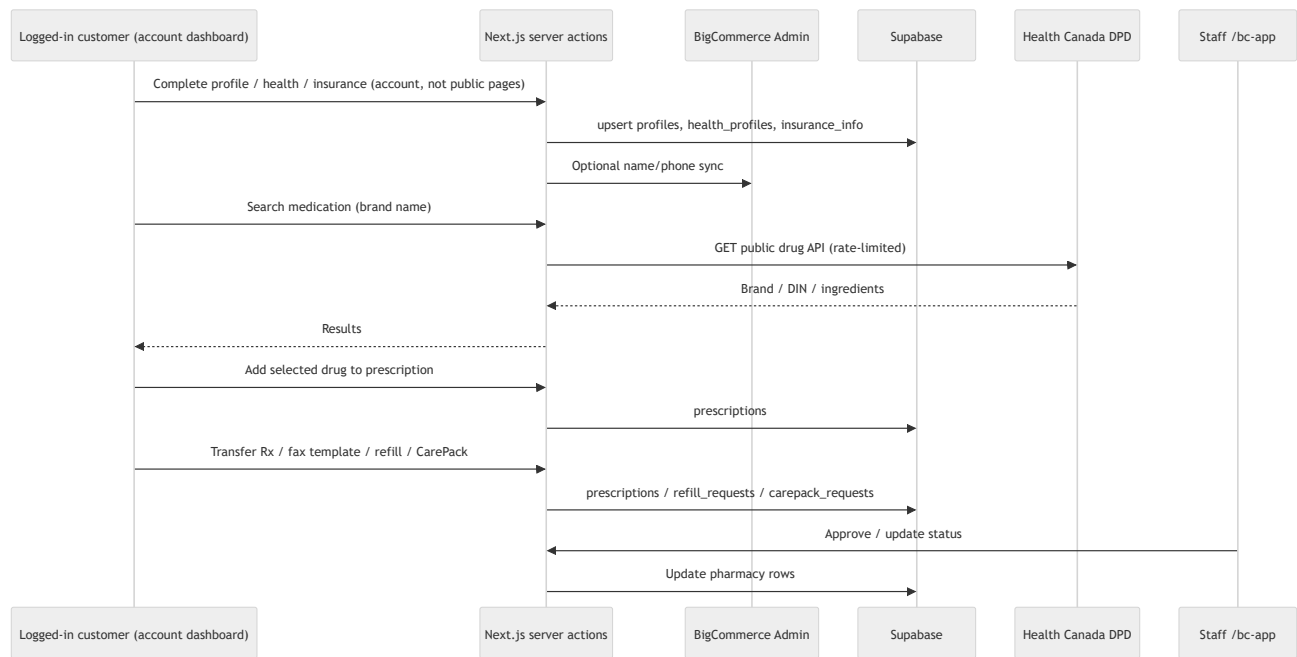
2. System context



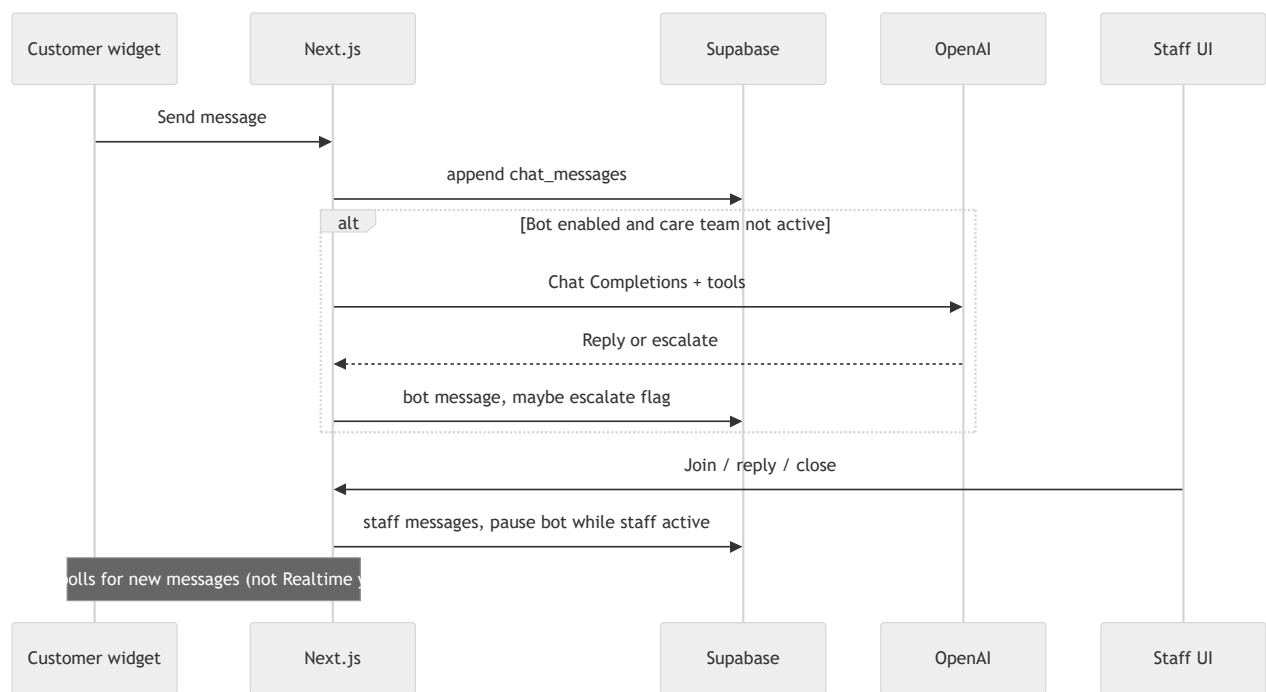
3. Data flow — commerce



4. Data flow — onboarding and pharmacy



5. Data flow — virtual care chat



6. Security controls (S1–S8)

Each item is a control so shopping, health data, staff access, and third-party APIs stay inside the intended boundaries.

ID	Topic	If left unchecked	Status	Control
S1	Who can query Supabase	High	In place	Row Level Security is on for public tables, with no anon/authenticated policies (deny by default). Browsers never receive a database key. Next.js uses the service role on the server only; that role bypasses RLS by design.
S2	Where health data lives	High	In place	Health profile, insurance, prescriptions, CarePack, and chat live in Canadian Supabase (ca-central-1) — not in BigCommerce. Patients add prescriptions by transfer or doctor fax only ; we do not collect Rx photos.
S3	Staff access	Med	In place	Staff work inside BigCommerce → Apps → Liivv Staff (/bc-app) , using the store's signed-in load token (BC app session). /staff is not a product surface (404).
S4	AI chat assistant	Med	Off	The assistant is off (VIRTUAL_CARE_BOT_ENABLED=false). If it were on, customer message text would go to OpenAI even when the reply is “talk to a pharmacist.” Human care-team chat in /bc-app stays in Supabase either way.
S5	Environment secrets	Med	In place	Secrets live in Vercel (encrypted) and in gitignored .env.local. Nothing privileged is committed. Rotate on personnel change.
S6	Embedded staff app frames	Med	In place	Staff work inside a BigCommerce control-panel iframe (/bc-app). Cookies used in that iframe are SameSite=None; Secure (partitioned where needed). CSP frame-ancestors is allowlisted, not open.
S7	Customer email	Low	In place	Password reset and most transactional mail are BigCommerce's . The storefront only triggers reset via GraphQL.
S8	Medication search (Health Canada DPD)	Low	In place	Customers look up licensed Canadian drugs through Health Canada's free public DPD API so they can add the right product to a prescription. Search is a server proxy , rate-limited at 60 requests / IP / minute , so Liivv is not an open relay to Health Canada. The selected medication is saved on the prescription in Supabase . Queries to DPD are catalog lookups — not a patient-data path.

Additional controls

- Privileged work runs in Server Components / server actions (tokens never held in the browser)
- Stripe and BigCommerce webhooks are verified
- BC app session is bound to the store hash
- Cart ID is inside a signed JWT

7. Authentication boundaries

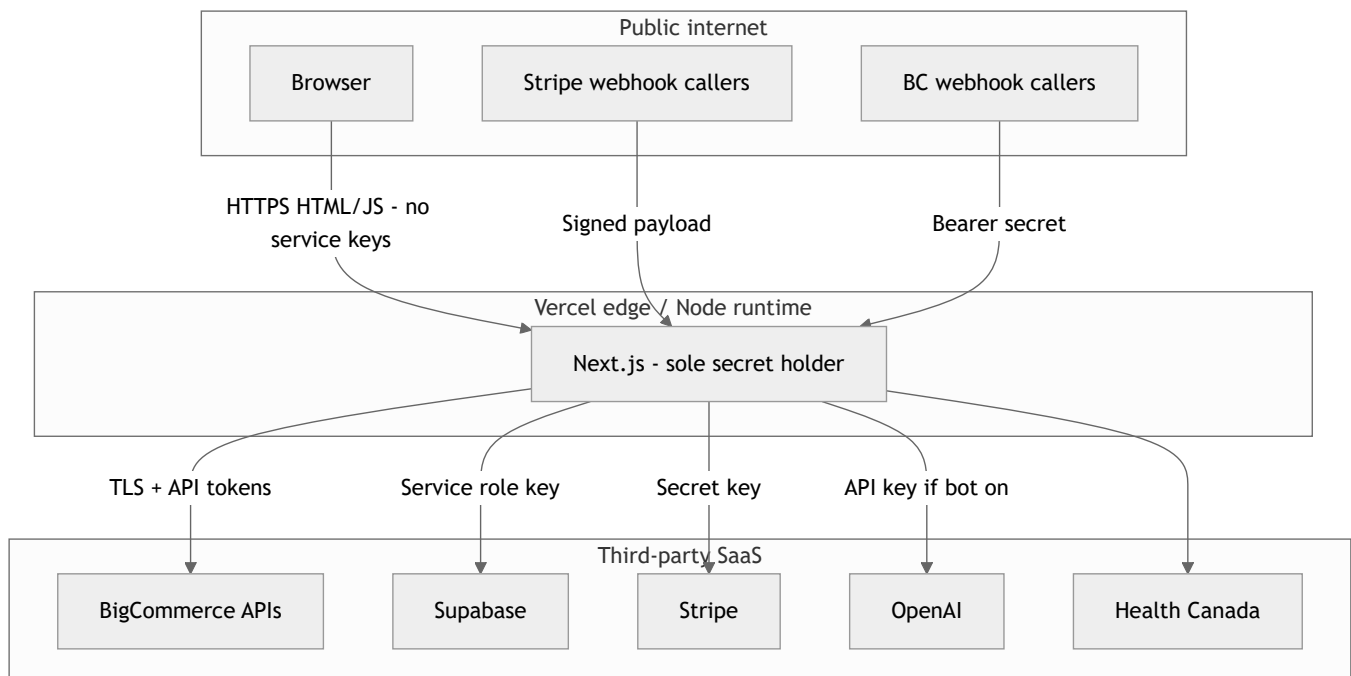
Plane	Mechanism	Cookie	Secret	Path / lifetime
Customer	NextAuth → BC GraphQL login or Customer Login JWT	Auth.js session JWT	AUTH_SECRET	Site-wide
Anonymous cart	Signed JWT containing cartId	authjs.anonymous-session-token	AUTH_SECRET	7 days
Staff (BC app)	OAuth install + signed load	liivv_bc_app	BIGCOMMERCE_APP_CLIENT_SECRET	/bc-app, 12 hours

ENABLE_ADMIN_ROUTE only redirects /admin to the BigCommerce control panel. It is **not** the pharmacy staff portal.

Webhooks

- Stripe: stripe.webhooks.constructEvent + STRIPE_WEBHOOK_SECRET
- BigCommerce: Authorization: Bearer + BIGCOMMERCE_WEBHOOK_SECRET

8. Network and trust boundary



Secrets inventory (server-only unless noted)

Secret	Purpose
AUTH_SECRET	Auth.js + anonymous cart JWT
BIGCOMMERCE_STOREFRONT_TOKEN	Storefront GraphQL
BIGCOMMERCE_ACCESS_TOKEN	Admin REST (orders, customers)
BIGCOMMERCE_CLIENT_ID / CLIENT_SECRET	Customer Login API JWT
BIGCOMMERCE_WEBHOOK_SECRET	Product → Stripe sync
BIGCOMMERCE_APP_CLIENT_ID / SECRET	Embedded staff app
STRIPE_SECRET_KEY / STRIPE_WEBHOOK_SECRET	Payments
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	Public — Stripe.js only
SUPABASE_URL / SUPABASE_SERVICE_ROLE_KEY	DB (bypasses RLS)
OPENAI_API_KEY	Virtual care bot (unused while flag is off)

9. Key evidence paths (code)

```
core/package.json
.env.example
core/auth/index.ts
core/lib/supabase/client.ts
core/lib/supabase/onboarding-schema.sql
core/lib/supabase/pharmacy-schema.sql
core/lib/bc-app-session.ts
core/lib/staff-access.ts
core/lib/content-security-policy.ts
core/lib/pharmacy/medication-rate-limit.ts
core/lib/stripe/webhook-handlers.ts
core/lib/virtual-care-bot/
core/app/api/medications/
core/app/api/stripe/webhook/route.ts
core/app/api/bigcommerce/webhook/route.ts
core/app/api/bigcommerce/app/{auth,load,uninstall}/route.ts
core/app/bc-app/
```

Companion walkthrough (if used): `liivv-it-architecture.canvas.tsx`.

Product-shaped summary: [Liivv-Architecture.md](#).